

Chapter

Energy Resources

1. (i) Mention two ways in which Hirakud dam has helped the people of Odisha?
(ii) Across which river is the Hirakud dam built? MAIN 2024

2. Give reasons for each of the following:

- (i) Geothermal energy is not popular in India.
(ii) Solar energy is gaining popularity.
(iii) Petroleum is considered a harmful source of energy. MAIN 2024

3. Name the following:

- (i) A mineral used to generate nuclear energy.
(ii) The non-conventional energy source that has the most widespread potential usage in India.
(iii) An offshore oil field of India. MAIN 2023

4. Identify the source of energy that leads to pollution: MAIN 2024

- (a) Tidal energy
(b) Wind energy
(c) Natural gas
(d) Geo-thermal energy

5. Hirakud dam is based on which of the following rivers? COMP 2024

- (a) River Godavari
(b) River Mahanadi
(c) River Krishna
(d) River Narmada

**6. Which types of resources of energy are coal, petroleum and natural gas?
SQP 2023**

- (a) Conventional sources
- (b) Non-renewable sources
- (c) Non-conventional sources
- (d) Both (a) and (b)

7. Tidal energy can be harnessed at ____ . MAIN 2022

- (a) Gulf of Kutch
- (b) Sundarbans
- (c) Andaman and Nicobar Islands
- (d) All of the above

8. Light distillates : gasoline :: middle distillates : ____ . COMP 2021

- (a) Diesel
- (b) Lubricating oils
- (c) Naphtha
- (d) LPG

9. Petroleum is used MAIN 2023

- (a) As a fuel
- (b) As raw material
- (c) For power generation
- (d) All of the above

10. Where are the tertiary coal fields located ? COMP 2021

11. Which coal fields supply coal to Rourkela Steel Plant ? MAIN 2023

12. Why is conventional source of energy not considered an ecofriendly source of energy? MAIN 2023

13. With reference to nuclear energy answer the following questions :

- (i) "It has vast potential for future development". Give reason.
- (ii) State any two sources of nuclear energy MAIN 2022

14. (i) Why do the natural gas deposits mostly accompany oil deposits ?
(ii) What is added to domestic gas cylinders, which gives a foul smell ? Why ?
COMP 2021

15. How is Gondwana coalfield different from tertiary coalfield in terms of location and quality? MAIN 2022

16. How is Bhakra Nangal dam beneficial for Punjab? MAIN 2023

17. Mention one advantage of using wind energy MAIN 2023

18. What is meant by irreplaceable and replaceable sources of power ?
MAIN 2021

19. Name one coal field each in the following States: Jharkhand, Orissa, West Bengal, Chhattisgarh. SQP 2023

20. Mention any two advantages of Nuclear power. SQP 2022

21. What are non-conventional sources of energy? SQP 2022

22. Give two advantages and one disadvantage of natural gas. MAIN 2023

23. Give a reason for each of the following :

(i) Conventional sources of energy are called the resources of the future.

(ii) India is an advantageous position in terms of solar energy.

(iii) Harvesting of solar energy does not require much maintenance.

MAIN 2022

Solution

1. (i) Two ways in which Hirakud Dam has helped the people of Odisha:

(a) **Irrigation** – Provides water for irrigation, enabling cultivation even during dry seasons.

(b) **Flood Control** – Helps control floods in the Mahanadi river basin, protecting nearby areas.

(ii) **River** – The Hirakud Dam is built across the **Mahanadi River**.

2. (i) **Geothermal energy is not popular in India** – India has limited geothermal resources, and tapping this energy requires advanced technology and high costs, making it less viable.

(ii) **Solar energy is gaining popularity** – It is renewable, abundant, and environmentally friendly. Advances in technology and decreasing costs have made it more accessible and sustainable.

(iii) **Petroleum is considered a harmful source of energy** – Its combustion releases greenhouse gases and pollutants, contributing to air pollution, global warming, and environmental degradation.

3. (i) Uranium.

(ii) Solar energy.

(iii) Bombay High.

4. (c) **Natural gas.**

Natural gas is a fossil fuel, and its combustion releases carbon dioxide and other pollutants into the atmosphere, contributing to air pollution.

5. (b) **River Mahanadi**

The **Hirakud Dam** is built across the **Mahanadi River** in Odisha.

6. (d) Both (a) and (b).

Coal, petroleum, and natural gas are **conventional** sources of energy and are **non-renewable**, so the answer is (d) Both (a) and (b).

7. (d) All of the above.

Tidal energy can be harnessed at **all the above locations** – Gulf of Kutch, Sundarbans, and Andaman & Nicobar Islands,

8. (a) Diesel

Correct. Middle distillates refer to petroleum products that are heavier than light distillates (like gasoline) but lighter than heavy oils. **Diesel** is a middle distillate.

9. (a) As a fuel

Correct. Petroleum is primarily used **as a fuel** for vehicles, industries, and heating purposes.

10. **Answer:** Tertiary coal fields are mainly located at **Makum in Assam and Palna in Rajasthan.**

11. The coal fields supplying coal to **Rourkela Steel Plant** are **Jharia, Raniganj, Talcher, and Korba.**

12. Conventional sources of energy are **not eco-friendly** because burning coal, oil, and natural gas releases **greenhouse gases, air pollutants, and toxic residues**, leading to **climate change, environmental degradation**, and the **depletion of finite resources.**

13. (i) **Reason for vast potential:** Nuclear energy can produce a **large amount of energy from a small quantity of fuel**. It is highly efficient and emits **minimal greenhouse gases**, making it a cleaner energy source suitable for meeting future energy demands.

(ii) **Two sources of nuclear energy:** **Uranium and Thorium**, which are found in **Jharkhand, Rajasthan, and Kerala.**

14. (i) Reason why natural gas deposits mostly accompany oil deposits: Natural gas is formed from the **decomposed remains of plants and animals** buried underground over millions of years, **just like petroleum**, so it is often found alongside oil deposits.

(ii) Substance added to domestic gas cylinders and reason: Ethyl mercaptan is added to LPG to give it a **foul smell**, making it **easily detectable and preventing accidents** since LPG is naturally odorless.

15. Gondwana coalfields are located in central and eastern India, such as Jharkhand, Odisha, and Chhattisgarh, whereas tertiary coalfields are found in northeastern India, mainly in Meghalaya and Assam. Gondwana coal is older and of higher quality, primarily **bituminous**, while tertiary coal is younger and of lower quality, mostly **lignite**.

16. The **Bhakra Nangal Dam** benefits Punjab by:

- 1. Providing irrigation water** – It supports agriculture by supplying water to the fields, helping farmers grow crops even during dry periods.
- 2. Generating hydroelectric power** – It produces electricity, contributing to the energy needs of Punjab and nearby regions.

17. One advantage of using **wind energy** is that it is a **renewable and clean source of energy**, producing no greenhouse gases or pollution, and is therefore environmentally friendly.

18. (i) Irreplaceable Sources of Power: These are energy sources that **cannot be replenished** once used. After consumption, they cannot be restored or replaced. Examples include coal, petroleum, and natural gas.

(ii) Replaceable Sources of Power: These are energy sources that **can be renewed or replenished** naturally or through human efforts after use. Examples include solar energy, wind energy, and hydroelectric power.

19.

State	Coal Field
Jharkhand	Jharia
Odisha	Talcher
West Bengal	Raniganj
Chhattisgarh	Korba

20. Two advantages of nuclear power are:

1. **High energy output** – A small amount of nuclear fuel can produce a large amount of energy.
2. **Low emissions** – Nuclear power generates minimal greenhouse gases, making it environmentally friendly.

21. Non-conventional sources of energy are **renewable and sustainable energy sources** that are environmentally friendly and naturally replenished. They are alternatives to fossil fuels and help reduce pollution and dependence on non-renewable resources. Solar energy, Wind energy, Biomass energy, Tidal energy, Geothermal energy.

22. Advantages of Natural Gas:

1. **Clean Energy:** Burns cleaner than other fossil fuels, producing fewer pollutants and lower carbon dioxide emissions.
2. **Efficient Energy Source:** Provides reliable energy for heating, electricity generation, and transportation.

Disadvantage of Natural Gas:

- It is a fossil fuel with finite reserves, making it unsustainable in the long term.

23. (i) **Conventional sources of energy are called the resources of the future** because they have been the primary sources of energy for centuries and

continue to meet the major energy needs of industries, transport, and households.

(ii) **India is in an advantageous position in terms of solar energy** because it receives abundant sunlight throughout the year, especially in regions like Rajasthan, Gujarat, and Madhya Pradesh, making solar power generation highly feasible.

(iii) **Harvesting of solar energy does not require much maintenance** because solar panels have no moving parts, so they operate silently and need only occasional cleaning to remove dust or debris.